EXPLAINER · 2026-08-20

GumBall6900

How GUM BALL 6900 turns community conviction into an onchain portfolio

A plain-language walkthrough of the whole protocol: what GBX is, what signaling does, where revenue comes from, how acquisitions and redemptions work, and what the design explicitly does not guarantee.

VERSION

2.0.0

DATE

2026-08-20

SOURCE COMMIT

dc67d7c4d634…

PROTOCOL STATUS

Development snapshot at the pinned commit. Describes the current core after ADR 0033, ADR 0034, and ADR 0035; not approved for user funds.

DEPLOYMENT

Not deployed on any network. No signed deployment manifest exists.

INDEPENDENT AUDIT

No independent external audit has been performed.

Before you read on: this protocol is not deployed, not audited, and not approved for user funds. This article describes what the code at commit `dc67d7c` does, not a live product. Nothing here is investment advice.

1. The central idea

Imagine an investment club with three unusual rules.

First, it has no manager. Members commit a token to express conviction — "we should be buying wrapped bitcoin" — and the club's income flows toward whichever holdings carry the most committed conviction. Move your commitment, and the flow follows.

Second, it never negotiates a purchase. When it has money to spend, it holds a public auction: it offers its cash to anyone willing to hand over the target asset and lowers its asking price steadily until someone accepts. No appraiser, no price feed, no discretion.

Third, leaving is arithmetic rather than a request. Burn your membership tokens and the vault hands you your proportional share of whichever holdings you name. No redemption window, no gate, no manager who can say no.

That is GUM BALL 6900. These are not policies people follow — they are smart contracts that execute. The manager is not honest; the manager does not exist.

2. Why ordinary funds require trust

Buying into a conventional fund means trusting a chain of promises: that the fund holds what the statement says, that the custodian really has the assets, that the fee calculation is right, that you can withdraw on the stated terms. Each is enforced by law and reputation — slowly, and after the fact.

Early onchain funds moved the assets on-chain but often kept the trust. Look closely at many "decentralized index" products and you find an admin key that can change holdings, an upgradeable contract that can change rules, a pause switch that can stop withdrawals, and an oracle that decides what everything is worth. Same promises, different paperwork.

GUM BALL 6900's premise is that if you remove *every* discretionary lever, what remains is either verifiable or absent. At this commit the protocol has no upgrade path, proxy, pause switch, rescue or sweep function, arbitrary-call executor, migration route, price oracle, NAV calculation, rebalancing engine, or keeper role. The treasury has no owner at all.

The honest flip side: removing every lever also removes every repair. A bug cannot be patched. A deployment mistake cannot be corrected.

3. What GBX represents

GBX is the protocol's token. It is an ordinary transferable ERC-20 — you can hold it, send it, and trade it.

It is created in exactly two ways. At deployment, the contract mints **20,000,000 GBX** once, and that allocation becomes the protocol's permanent market liquidity (more on that in section 11). After that, all new GBX comes from **mining** — continuous issuance to whoever occupies the mine's slots.

The crucial structural fact is that mint authority is handed to one contract, exactly once, at deployment, and then permanently locked. There is no way to add a second minter, replace the first, or reopen the handover. Burning GBX doesn't reopen it either. Supply reconciles exactly: total supply always equals everything ever minted minus everything ever burned.

There is **no supply cap**. Issuance halves as cumulative mining crosses fixed thresholds, but it settles at a permanent floor that is strictly positive rather than falling to zero. GBX inflates forever, slowly.

Holding GBX gives you two rights, and it is worth being precise about which is which:

- **Signal with it**, and you direct protocol revenue and earn rewards.
- **Burn it**, and you withdraw your share of the treasury.

Holding GBX in your wallet does neither of those things by itself.

4. What it means to signal

The protocol has **Strategies** — each one a standing mandate to acquire one particular asset. A Strategy for wrapped bitcoin. A Strategy for a staked-ETH token. And so on.

Signaling is a single atomic step with three parts that always happen together:

1. You deposit GBX into **SignalGBX**, ticker **sGBX**.
2. You receive exactly the same amount of sGBX.
3. That sGBX is committed to one Strategy you name.

Withdrawing reverses all three at once. There is deliberately **no in-between state**: you cannot hold sGBX that isn't committed to something. Every sGBX unit in existence is assigned to exactly one Strategy at all times.

SIGNALING IS ONE STEP, NOT THREE

All of this happens in a single transaction, or none of it happens.

Deposit GBX
held by the receipt contract

→

Receive sGBX
same amount, non-transferable

→

Assign to a Strategy
named at the same moment

one transaction · withdrawing reverses all three

There is deliberately no state in between. You cannot hold sGBX that is not pointed at something, so every unit of signal that exists is a unit of signal doing work.

DEPOSIT, RECEIPT, AND COMMITMENT ARE ONE ATOMIC STEP. THE ABSENCE OF AN IN-BETWEEN STATE IS WHAT MAKES THE AMOUNT OF SIGNAL THAT EXISTS AND THE AMOUNT DOING WORK THE SAME QUANTITY.

That is a real design choice with consequences, so it's worth stating plainly. It means every unit of signal that exists is a unit of signal doing work — there is no parked, idle, or purely symbolic position. The amount of sGBX in existence and the amount actually directing revenue are the same number, always.

sGBX is **non-transferable**. You cannot send it, sell it, or lend it. Any transfer attempt reverts.

You can spread your position across Strategies by signaling each one separately — 700 GBX to the bitcoin Strategy, 300 to the staked-ETH Strategy. These are absolute amounts, not percentages.

Three properties matter:

- **There is no lock-up, cooldown, or epoch.** You can signal, move, and withdraw in consecutive blocks.
- **You can always withdraw.** Withdrawal is bounded only by what you actually have committed to the Strategy you're withdrawing from.
- **Every signal change first settles the revenue that accrued under the old weights.** Moving your signal never claws back or redirects money that already accrued to someone else. It only affects flow from that moment on.

The complete user surface is four functions: signal, signal using a gasless approval on the underlying GBX, move signal between Strategies, and withdraw signal.

5. What sGBX is, and what it is not

sGBX is a **receipt**, not a second tradeable token. It exists to record two things: that you deposited GBX, and which Strategy you pointed it at. It cannot leave your wallet, and it has no market.

So there are two positions a person can hold, and they are not interchangeable:

POSITION	DIRECTS REVENUE?	EARNS STRATEGY REWARDS?	CAN REDEEM FROM THE FUND?
Holding liquid GBX in a wallet	No	No	Yes (by burning)
Signaling a Strategy with sGBX	Yes	Yes	No (withdraw first)

There is one more thing sGBX quietly does. It keeps **vote checkpoints** — a standing record of how much sGBX each account held at each past block, in the standard format governance systems read. Your first signal switches that recording on automatically, with no second transaction.

Nothing in the protocol reads those checkpoints today. They are there because the protocol expects an external governance system to be attached later, and that system will need them. Which system, and on what rules, is not decided — §13 is honest about what that means.

One property worth understanding now, because it does not go away: **checkpoints survive withdrawal.** Once a block has passed, the record of what you held at that block is permanent history, even if you withdrew everything afterwards. If a governance system is later attached that grants power based on past blocks, someone could borrow GBX, signal it, let a block pass, withdraw, and still carry the recorded weight. Whether that is exploitable depends entirely on the system chosen. It is tracked internally as finding G-01 and is one of the reasons that choice has not been rushed.

6. Where the money comes from

Protocol revenue arrives in **USDG**, a stablecoin the protocol neither issues nor controls. In the intended deployment it has six decimal places, which matters later for the arithmetic.

There are two revenue sources, and only two.

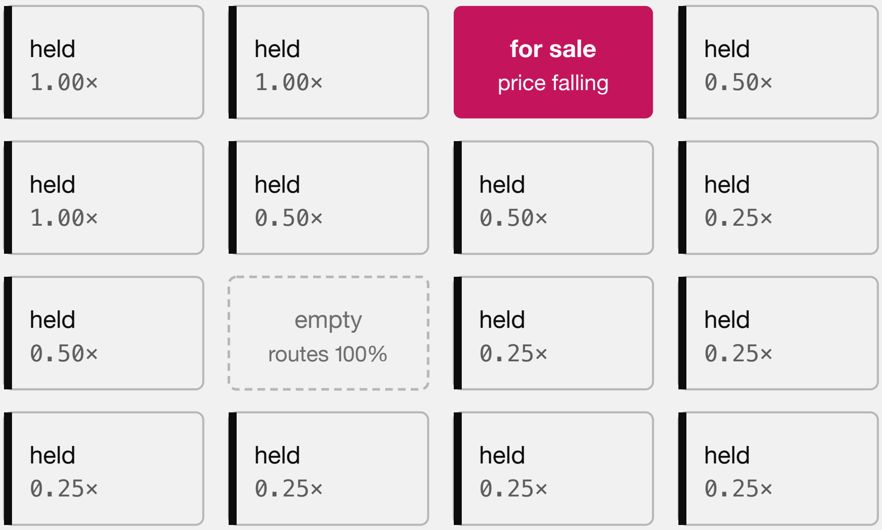
Mining. The mine has exactly sixteen permanent slots. Whoever occupies a slot accrues GBX continuously at a fixed rate, minted when that slot next changes hands. To take a slot, you win its auction: the price to displace the current occupant starts at some level and falls in a straight line to zero over one hour, then sits at zero until someone takes it.

When you take an **occupied** slot, 80% of what you pay becomes a claim for the miner you displaced, and 20% becomes protocol revenue. When you take an **empty** slot, there is nobody to compensate, so 100% becomes protocol revenue. There is no team fee anywhere in this. The displaced miner's 80% is held as a claim they withdraw when they like — anyone can trigger the withdrawal, but the money can only ever go to the miner.

Two things a prospective miner should understand. First, your GBX rate is **locked for your entire tenure** — halving issuance, redemptions, and other slots' handoffs never change it. Only a newly occupied or replaced slot receives the current global TPS divided by sixteen. Second, less comfortably: **the 80% handoff is not guaranteed.** You receive it only if someone later replaces you at a nonzero price, and since the price falls to zero after an hour, a successor can replace you having paid nothing. You keep the GBX you accrued, but no handoff payment. Interfaces must not present that 80% as principal, yield, or a refund.

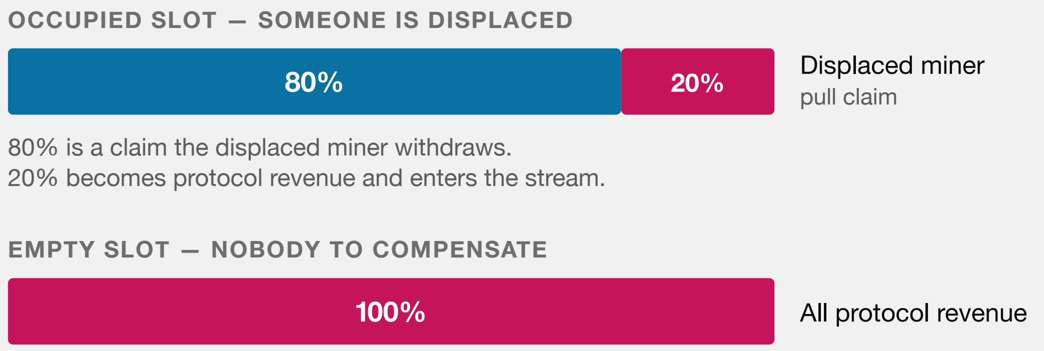
16 PERMANENT SLOTS

fixed at construction · no owner · no capacity setter



Each slot runs its own auction, falling to zero over 1 hour, and pays 80% to whoever it displaces.
A slot's issuance rate is fixed when it is taken and never moves again. The mixed rates above are incumbents who took their slots under different halvings.

THE MINING MARKET IS A FIXED BOARD OF SIXTEEN INDEPENDENT SLOTS. NOTHING CAN ADD A SEVENTEENTH, AND NO SLOT'S RATE CAN BE CHANGED BY ANYONE ONCE IT IS TAKEN.



MINING HANDOFF. TAKING AN OCCUPIED SLOT COMPENSATES THE MINER YOU DISPLACED; TAKING AN EMPTY ONE FUNDS THE PROTOCOL ENTIRELY.

Liquidity fees. The 20,000,000 GBX genesis allocation sits in a single Uniswap v4 position pairing GBX with USDG, held permanently in a contract with no owner and no withdrawal function. Anyone may call a public function that collects its trading fees: the USDG becomes revenue, the GBX is burned, and the underlying liquidity is verified unchanged. There is no reward for the caller, so fees can sit uncollected until someone volunteers the gas.

7. How Resonance directs revenue over time

Revenue does not get handed out the moment it arrives. It flows into a contract called **Resonance**, which releases it as a **rolling seven-day stream**.

Think of it as a tank set to drain evenly over seven days. Whatever comes out at each moment is divided among Strategies in proportion to the sGBX signaling them *at that moment*.

Two design details exist to prevent specific attacks.

Revenue waits in a router until it is worth restarting the stream for. New revenue accumulates in a staging contract called **ResonanceRouter**, which forwards its balance only once that balance is at least as large as the exact amount still left in the current schedule. Once it qualifies, the router forwards *everything*, and Resonance combines the new money with the old remainder and restarts a fresh seven-day stream. So restarting the stream early is possible but expensive — you must match what's left. It also means a mining payment can sit in the router for a while before it appears in the stream; interfaces must show those as different states.

Streaming is lazy. Entitlement accrues with time in the arithmetic, but tokens move only when someone triggers a settling action — a signal change, a revenue notification, a payout, or a purchase. So a Strategy's visible token balance understates what it is owed, and any interface that reads the raw balance and calls it inventory is wrong.

8. A numeric walkthrough: how signal allocation works

Let's make this concrete. All figures below are exact integer arithmetic at six decimals for USDG and eighteen for sGBX.

Setup. Resonance receives a notification of **604,800 USDG**. Seven days is 604,800 seconds, so the stream rate works out to exactly **1 USDG per second** with no remainder. (When the division isn't clean, the leftover raw units are paid out one extra unit per second at the start of the period — nothing is discarded.)

Three participants and three Strategies:

PERSON	SIGNALS	SGBX MINTED
Ana	1,000 GBX to **Strategy A** (wrapped bitcoin)	1,000
Ben	3,000 GBX to **Strategy B** (staked ETH)	3,000
Cara	600 GBX to **Strategy C** (a governance token)	600

Total active signal weight: 1,000 + 3,000 + 600 = **4,600 sGBX** — which is also the entire sGBX supply, since none of it can sit idle.

Day 1 streams 86,400 USDG (86,400,000,000 raw units), split by weight:

STRATEGY	WEIGHT	SHARE	DAY 1 USDG
A	1,000	5/23	18,782.608695
B	3,000	15/23	56,347.826086
C	600	3/23	11,269.565217
			86,399.999998

Notice the total: **0.000002 USDG short**. That residue is the floor of the integer division. It is not paid to anyone, does not go to the treasury, and stays in Resonance permanently. This is a deliberate, documented trade-off — section 15 returns to it.

Day 2. Ben decides staked ETH is the wrong bet and moves his entire 3,000 signal from Strategy B to Strategy A. Crucially, this settles Day 1 first — Ben's move does **not** claw back the 56,347.826086 USDG that already accrued to Strategy B. From now on:

STRATEGY	WEIGHT	SHARE	DAY 2 USDG
A	4,000	20/23	75,130.434782
B	0	0	0
C	600	3/23	11,269.565217

Day 3. Cara withdraws her 600 signal from Strategy C — which in one atomic step removes the signal, burns her 600 sGBX, and returns her 600 GBX. Active weight is now just Ana's 1,000 plus Ben's 3,000, both on Strategy A:

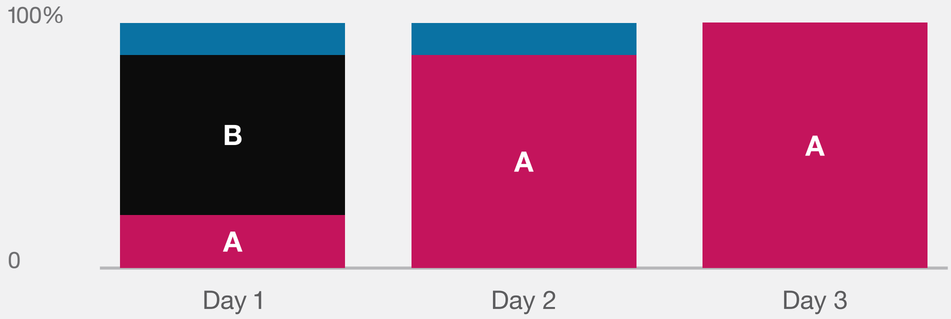
STRATEGY	WEIGHT	SHARE	DAY 3 USDG
A	4,000	100%	86,400.00

Totals after three days:

STRATEGY	ACCUMULATED USDG
A	180,313.043477
B	56,347.826086
C	22,539.130434

WHERE ONE DAY OF REVENUE GOES

Each day is the same 86,400 USDG, split by whoever is signaling at the time.



Day 2: Ben moves to A — Day 1 is already settled, so his move changes only later flow.
Day 3: Cara withdraws entirely, so A takes everything.

THE WORKED EXAMPLE FROM THE TEXT: THE SAME DAILY REVENUE, REDIRECTED AS SIGNALERS CHANGE THEIR MINDS.

Four days and 345,600 USDG remain in the stream. Cara earned Bribe rewards on Strategy C for two days and stopped the moment she withdrew. Ben's Strategy B holding is frozen at what it earned — it will accumulate nothing further unless someone signals it again.

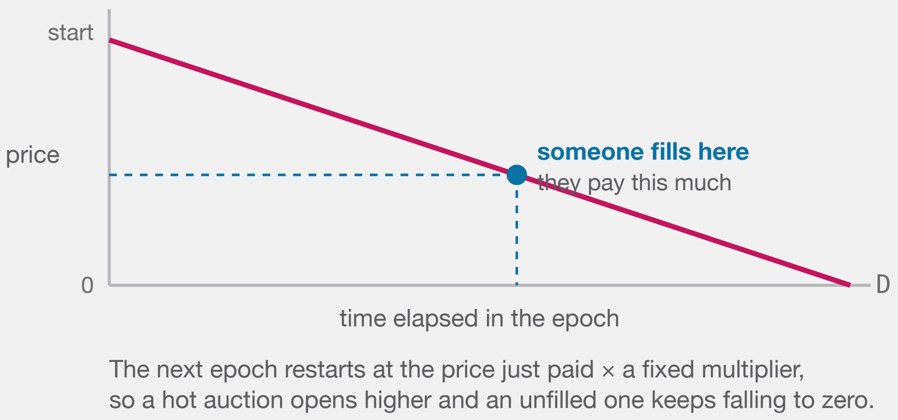
9. What a Strategy auction does

A Strategy now holds USDG and needs to convert it into the asset it is mandated to acquire. It does this with a **descending-price auction** — the design is sometimes called a "reverse Dutch auction" after the lineage it derives from; mechanically, the price starts high and falls until somebody takes the trade.

Here is the whole mechanism. The Strategy offers its **entire** USDG balance. It asks to be paid a certain quantity of the target asset. That asking quantity falls in a straight line to zero over a configured period (at minimum one hour, at maximum one year). Anyone may fill it at any moment by paying the current asking price. Once filled, the next auction starts at the price just paid multiplied by a fixed multiplier — so a hot auction restarts higher — subject to a floor and a ceiling.

There is no oracle. The auction *is* the price discovery. If the Strategy is asking too much, nobody fills it and the price falls. If it is asking too little, someone fills it immediately and the next auction starts higher.

Buyers are protected by three parameters on their own transaction: the auction round they expect (so a competing fill can't change the price under them), a deadline, and a maximum they will pay.



BOTH THE MINING SLOTS AND THE STRATEGY AUCTIONS USE THE SAME SHAPE: THE ASKING PRICE FALLS IN A STRAIGHT LINE TO ZERO, AND WHOEVER THINKS IT IS CHEAP ENOUGH FILLS IT.

10. A numeric walkthrough: acquisition and redemption

Continuing the example, **Strategy A** holds **180,313.043477 USDG** and is mandated to acquire wrapped bitcoin.

The auction. Strategy A's auction runs over 24 hours with a starting price of **4 WBTC**. The price falls linearly. A buyer watching decides that 180,313 USDG is worth roughly 2 WBTC and waits.

At 13.2 hours elapsed (47,520 of 86,400 seconds, so 55% of the way through), the price has fallen 55%:

```
price = 4 WBTC − (4 WBTC × 47,520 ÷ 86,400) = 4 − 2.2 = 1.8 WBTC
```

The buyer fills. They pay **1.8 WBTC** and receive **180,313.043477 USDG**. (The purchase first pulls any revenue released to the Strategy through that exact timestamp, so they receive everything the Strategy is owed, not just its visible balance.)

Settlement — the 90/10 split. The 1.8 WBTC is immediately divided by the protocol's one bounded rule:

```
Fund   90%  →  1.62 WBTC   (treasury backing for every GBX holder)
Bribe  10%  →  0.18 WBTC   (reward for Ana and Ben, who signaled Strategy A)
```

Each share is recorded as a separate liability and delivered by its own call that anyone can make. This matters: if the treasury or the reward pool has a problem with that token, **only that leg stalls**. The other leg still settles, and neither can freeze the auction itself.

EVERY ACQUIRED PAYMENT

90%	10%
Fund treasury backing for every GBX holder	**Signalers** reward for signaling this Strategy

Fixed in code. No setter, no governance parameter, no caller-chosen destination.

EVERY ASSET THE PROTOCOL ACQUIRES IS DIVIDED BY AN IMMUTABLE RULE THE MOMENT IT ARRIVES.

The split is also **cumulatively exact**. It carries the fractional remainder between payments, so a buyer cannot starve the reward share by paying in tiny increments. Ten separate one-unit payments produce exactly 9 units to the Fund and 1 to the Bribe — not zero to the Bribe ten times over.

The next auction round then starts at 1.8 WBTC × the configured multiplier — at 1.5×, that is **2.7 WBTC**.

So Ana and Ben are paid directly out of the acquisition. They split 0.18 WBTC in proportion to their signal — Ana 1,000/4,000 and Ben 3,000/4,000, so 0.045 and 0.135 WBTC respectively. On top of that they may also earn separately funded **Bribes** — see section 12.

Redemption. Time passes. Many auctions run. Suppose the Fund now holds **50 WBTC** and **400 ETH**, GBX total supply is **100,000,000**, and miners have accrued **1,000,000 GBX** that has not yet been minted.

Diego holds 250,000 GBX and wants out. He calls redeem, naming WBTC and ETH.

Step 1 — pending mining counts first. Before doing anything else, the Fund reads Mine's effective supply: 100,000,000 minted GBX plus 1,000,000 accrued unminted GBX, or **101,000,000 GBX**. This constant-time view does not mint or touch every slot, and stops a redeemer from ignoring GBX the miners have already earned.

Step 2 — one snapshot for everything. The Fund records supply-before-burn = 101,000,000 GBX and records its balance of each named asset. Every payout uses this same denominator.

Step 3 — the payouts.

```
WBTC: floor(50 WBTC × 250,000 ÷ 101,000,000) = 0.12376237 WBTC
ETH:  floor(400 ETH × 250,000 ÷ 101,000,000) = 0.990099009900990099 ETH
```

Step 4 — the burn. Diego's 250,000 GBX is burned. Supply falls to 100,750,000 GBX. Everything above happens in one atomic transaction: if any transfer fails, the entire redemption including the burn is undone.

REDEEMING

Burn your GBX → **Name the assets** you want → **Your slice**

For each asset you name:

`floor( Fund's balance × your GBX ÷ total GBX supply )`

The supply is snapshotted after every miner is credited, so nobody is diluted by unminted rewards. Assets you leave out are forfeited — permanently, to everyone else.

It is one atomic step. If any transfer fails, the burn is undone too.

REDEMPTION IN FULL. NO QUEUE, NO GATE, NO DISCRETION — ONLY ARITHMETIC AGAINST A SNAPSHOT.

Two things worth noticing. If Step 1 ignored pending mining, Diego's WBTC payout would have been 0.125 WBTC instead of 0.12376237 — the difference is exactly the dilution from recognising the miners' claim, and it is correct. And if the Fund had been holding, say, 2,000,000 GBX from a GBX-denominated auction that nobody had burned yet, that GBX would have counted in the denominator and reduced Diego's payout. Anyone can burn Fund-held GBX at any time, and a redeemer should do so first.

11. What the Fund holds

The Fund is an **ownerless raw-token treasury**. Not a curated portfolio — a treasury.

It has no administrator, no roles, and no asset registry. Assets can leave in exactly two ways: a GBX holder redeems, or someone burns Fund-held GBX. There is no sweep, rescue, recovery, or migration function of any kind.

The consequence of having no registry is that **anyone can send any ERC-20 to the Fund and it becomes redeemable backing**, reviewed or not. Being in the Fund does not make a token official. Official membership means having a registered Strategy — nothing more. That registered set is the closest thing the protocol has to an index: it is the curated list of assets the protocol is trying to acquire. But it is only a list. There are no target weights, no rebalancing, and no valuation — whatever proportions the Fund ends up holding are an outcome, not a plan. Interfaces must label unsolicited Fund balances separately.

Because you name the assets you want, you can skip a broken or worthless token rather than have it block your redemption. The trade-off is that **anything you skip is permanently forfeited** — it stays in the Fund for the remaining supply. There is no partial-claim ledger.

12. How signalers are rewarded

Signalers have two stacked income sources, and they arrive through the same contract.

First, the automatic 10%. Every time their Strategy acquires an asset, a tenth of that asset is streamed to whoever is signaling it, split in proportion to signal. This requires no external party and no negotiation — it is simply what the protocol does with every acquisition. It also aligns the incentive neatly: you earn the asset you voted to accumulate, so signaling for something worthless pays you in something worthless.

Second, Bribes. A "bribe," here, is not corruption — it is the standard term for an open, permissionless reward stream used to attract attention. Anyone can deposit additional reward tokens into a Strategy's pool, which streams them to that Strategy's signalers the same way.

Who would do that? A project that wants the protocol to accumulate its token; a DAO that wants its asset in a long-term treasury; a market maker who benefits from the auction flow. The protocol takes no position on motive.

Some deliberate details, which apply to both sources:

- **At most eight reward tokens per Strategy**, hard-coded, append-only. The cap exists so that entering, leaving, and settling a signal position stay bounded in gas no matter what anyone registers.
- **You cannot disturb someone else's live stream.** Adding rewards while a stream is running queues them behind it rather than restarting or slowing it. (This is the opposite of Resonance's behavior — the two are genuinely different mechanisms.)
- **If everyone stops signaling, the stream pauses** rather than draining with nobody to pay, and resumes when signal returns.
- **You can claim one token at a time**, so a single frozen reward token cannot block the rest.
- **Rewards you arrived too late for are not yours.** Fractional amounts that cannot be fairly assigned are routed to the Fund rather than redistributed to whoever happens to still be signaling.
- **Each reward token has a lifetime ceiling on how much can ever be streamed through it.** The number is astronomically large — for a normal eighteen-decimal token, far more than any real supply — and exists purely so that a token with an absurd unit count cannot be used to jam the reward arithmetic and trap other people's deposits. Reaching it would block further funding of that one token only; existing rewards, claims, and withdrawals keep working.

13. How governance works — and what is still missing

Start with the good news, because it is the larger part. There are exactly **four things** about this protocol that anyone can ever change:

1. Add a Strategy.
2. Permanently retire a Strategy.
3. Register a Bribe reward token (within the eight-token cap).
4. Set the signalers' share of each acquisition, anywhere from 0% to a hard ceiling of 20%.

NO OWNER
Nothing can be changed by anyone, ever.

Mine	sixteen slots, fixed
Fund	the treasury itself
LiquidityPosition	the v4 position
GBX	minter locked once
Strategy	auction parameters fixed
BribeRouter	90/10 hard-coded

ONE OWNER
Resonance, and only these three calls:

— add a Strategy
— retire a Strategy
— register a Bribe reward token

The holder of that address is
not yet chosen.

There is no upgrade path, pause switch, sweep, or migration route anywhere in the protocol — so this map is the complete authority surface, not a summary of it.

THE COMPLETE AUTHORITY SURFACE. ONE CONTRACT HAS AN OWNER; THE REST CANNOT BE ALTERED BY ANYONE. WHO HOLDS THAT OWNER ADDRESS IS THE PROTOCOL'S LARGEST OPEN QUESTION.

That is the complete list. All four live on **Resonance**, and Resonance is the only contract in the protocol that has an owner at all. The Mine has no owner. The Fund has no owner. The liquidity position has no owner. Nobody — not a developer, not a voter, not a future administrator — can change mining prices, issuance rates, halving parameters, the tail rate, mint authority, Fund assets, liquidity custody, the auction mechanism, or the sixteen-slot count. There is no upgrade path, no pause switch, and no sweep function to add one later.

The fourth item is the one genuine economic dial, and it is deliberately fenced. The reward share starts at 10% and can never exceed **20%**, so at least 80% of every acquisition reaches the treasury no matter who holds the owner address. A change applies only to purchases settled after it — it cannot reach back and reclassify an amount already recorded, a reward already streaming, or anything already in the Fund.

There is one guard rail on retirement: **the final live Strategy cannot be retired.** If only one remains, the call reverts. To replace it, the replacement must be added first. This guarantees there is always somewhere valid to signal, which matters because signaling is the only way to hold sGBX at all.

The part that is not finished

Those three actions sit behind a **single owner address** on Resonance. Who holds that address has not been decided.

The protocol used to ship its own voting contract and its own timelock. Both were removed (ADR 0034), because the intended deployment will hand ownership to an established external governance system instead — and maintaining a second, home-grown voting stack that was never going to be deployed meant carrying security surface for nothing. Removing it was the right call. But it does mean that today the protocol contains **no voting rules, no quorum, no proposal filter, and no execution delay of its own**. What sGBX offers is the vote checkpoints described in §5, sitting ready for a system that has not yet been attached.

Concretely, at this commit: whoever holds the Resonance owner address can add a Strategy, retire a Strategy, or register a reward token immediately — no vote, no waiting period, no way for anyone to object. They can also hand that address to someone else, or throw it away permanently. In development that address is simply the deployment fixture.

Four honest limitations follow:

- **Governance is a placeholder, not a mechanism.** Until an external system is chosen, reviewed, and actually given ownership, there is no meaningful answer to "who decides." Deploying without finishing that step would produce a protocol with an ordinary admin key — precisely the thing this design exists to avoid.
- **The protocol guarantees no delay, no veto, and no cancellation.** Whatever protections eventually exist will be properties of the external system, not of this code, and they will need reviewing on their own terms.
- **Retiring a Strategy is irreversible**, which makes both points above materially more serious.
- **Vote checkpoints survive withdrawal** (§5), so any system attached later must decide deliberately how much weight to give historical balances.

14. What happens when something fails

The protocol is built so a failure in one place doesn't cascade:

- **A broken token cannot trap your position.** Withdrawing signal is pure accounting — it never transfers a reward, payment, or revenue token, so a frozen third-party token cannot block them.
- **A frozen treasury cannot block an auction.** Auction payments are recorded as a liability and delivered by a separate, retryable call.
- **A broken asset cannot block redemption for everyone else.** You name what you want, so you can omit it.
- **A misbehaving token fails loudly.** Every transfer checks that the sender was debited and the receiver credited by exactly the requested amount, so fee-on-transfer and rebasing tokens revert rather than lose value silently. This is fail-closed evidence — it does *not* make an adversarial token safe.
- **Failed payouts stay owed and retryable.** A blocked recipient cannot redirect a liability elsewhere.

One failure mode has no recovery. When a Strategy is retired, its reward pool stays open for existing signalers but nobody new can join. If the **last** signaler withdraws while rewards remain, those rewards are stranded permanently — possibly an entire unvested stream, not merely dust. There is deliberately no refund or rescue. Interfaces must warn the final signaler before they exit.

15. What GUM BALL 6900 does not guarantee

Read this section twice.

- **It does not guarantee that assets appreciate.** The protocol buys what signalers point it at. That aggregates conviction; it does not make conviction correct.
- **It does not guarantee investment returns.** There is no yield, distribution, or promise of any kind. Your redemption is a share of whatever the Fund happens to hold, which may be worth less than you paid.
- **It does not guarantee auction liquidity.** An auction fills only if someone chooses to fill it. A price falling to zero guarantees a *price*, not a *buyer*.
- **It does not guarantee good signal choices.** There is no quality filter beyond whoever approves a Strategy, no diversification requirement, and no risk framework.

- **It does not guarantee safe external tokens.** USDG and every payment and reward token are third-party contracts that may be upgradeable, pausable, blocklisting, or malicious. The Fund accepts any ERC-20 without review.
- **It does not guarantee permanent frontend availability.** The contracts are permissionless, but no website, indexer, or API is guaranteed to exist. You may need to interact with contracts directly.
- **It does not eliminate smart-contract risk.** The code has not been independently audited, and because nothing is upgradeable, a discovered bug cannot be patched.
- **It does not eliminate governance risk.** The system that will own Resonance is unselected, so capture, delay, veto, and accountability are all open questions rather than settled properties (section 13).
- **It does not eliminate chain, MEV, or timing risk.** Auctions and mining slots are competitive, publicly visible opportunities. Transactions can be reordered, front-run, or censored, and chains can reorganize.
- **It does not eliminate regulatory risk.** Legal treatment in any jurisdiction is unresolved.
- **And it accumulates permanent dust.** Rounding residue and revenue streamed while nobody signals accrue in Resonance forever with no recovery path. Precision keeps individual amounts tiny, but **no lifetime bound on the total is claimed**.

16. Major risks, summarized

RISK	WHAT IT MEANS
No independent audit	No third party has reviewed this code. The single largest unknown.
Immutability	No patch, no pause, no rescue. A bug or deployment error is permanent.
Deployment correctness	Parameters, pool configuration, and role setup must be right the first time, forever.
Unresolved economics	Mining rate, halving threshold, tail rate, and price parameters not yet selected or modelled.
Unfinished governance	The external owner of Resonance is unselected; today one address holds all four powers outright.
Miner rollover	The 80% handoff arrives only if a successor pays. It can be zero.
Abandoned rewards	A retired Strategy's last signaler can strand an unbounded amount of rewards.
Accepted dust	Rounding and zero-signal intervals accumulate unrecoverable USDG in Resonance.
Third-party tokens	USDG and every acquired asset carry independent freeze, upgrade, and solvency risk.
Legal and provenance	Upstream code lineage and license reconciliation are unresolved release blockers.

17. Current project status

To be exact about where this stands at commit `dc67d7c`:

- **Not deployed.** No contract is live on any network. No signed deployment manifest exists. The intended target chain and the canonical USDG and Uniswap v4 addresses remain unresolved candidates.
- **Not audited.** No independent external audit has been performed, and symbolic analysis and formal verification have not been completed.
- **Internally tested.** At this commit the default test suite passes 329 tests and the integration suite passes 18, with zero failures and zero skips. That includes 29 stateful invariants each run 1,000 times to a depth of 500 transitions, with zero handler reverts. Hardhat bytecode parity, the SDK, the subgraph, the independent

TypeScript and Python economic simulations, documentation generation, linting, and type checking all pass. Real Uniswap v4 fee harvesting is exercised.

- **Static analysis, mutation testing, and external fuzzing are pinned to an older tree.** Those campaigns — Slither, Aderyn, Semgrep, Gitleaks, a Medusa run of 101,602 calls, an Echidna run of 100,213 calls, and a 43-mutant campaign that killed every mutant — were last executed before the governance removal (ADR 0034) and the Bribe reward cap (ADR 0035) landed. They remain useful engineering history. They are **not** current evidence for the code described in this article, and re-running them is an open task.
- **Four High-severity release gates remain open.** Two concern deployment: production mining and pricing parameters have not been selected or independently modelled, and no signed manifest yet proves the deployed bytecode, constructor arguments, and dependency addresses. Two concern governance: the external system that will own Resonance is unselected, and its voting, delegation, and delay semantics therefore remain unreviewed. Also outstanding: a second external-fuzzer seed, a monitored testnet rehearsal, and release review.
- **Legal and provenance clearance is outstanding.** The protocol's contracts adapt several upstream codebases whose licensing chain has not been resolved, including one with a GPL ancestor.

Nothing in this article should be read as a claim that the protocol is safe, audited, live, launched, production-ready, or suitable for funds you cannot afford to lose.

See also: the one-pager for a two-minute summary, and the technical whitepaper for exact formulas, state machines, accounting identities, and the full threat model.